

A theory of glassy dynamics based on interacting strings of hopping motions

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The dynamics of glassy systems are known to be dominated by string-like motions each consisting of a line of particles displacing one and other. We show using molecular dynamics simulations that these motions become highly repetitive back and forth motions close to the glass transition temperature and do not necessarily contribute to net displacements. We propose that structural relaxations are then brought about by pair-interactions between strings. A mean-field theory predicts an ideal bifurcation transition between a mobile phase with interacting strings and a frozen phase with isolated strings. Further variational calculation shows that realistic glassy systems assume the mobile phase close to criticality for an extended temperature range. We apply the theory to account for the viscosity of a common polymer exhibiting the Vogel-Fulcher-Tammann empirical temperature dependence.

I. INTRODUCTION

Many liquids can be quenched into a glassy state when cooled sufficiently rapidly [1, 2]. Widely studied examples include silicate glasses, polymers, and metallic glasses. The glass transition is one of the most puzzling phenomena in condensed matter physics. Fundamental questions including the nature of the transition and the mechanism for the dramatic slow-down of the dynamics remain unanswered. The glass transition occurs over a temperature range in contrast to many other familiar phase transitions. It is not known if there exists an underlying but avoided true phase transition. Theoretical approaches including the Adam-Gibbs theory [3], the mode-coupling theory [4], spin-glass related theories [5], and elastic models [6] provide important insights but remain controversial. Molecular dynamics (MD) simulations allow access to the microscopic details of the dynamics [7, 8]. A remarkable discovery is string-like motion in which neighboring particles arranged in a line displace one another from head to tail [9]. They have been observed not only in simulations of binary Lennard-Jones liquids [9, 10], monoatomic Dzugutov liquids [11], and bead-spring polymers [12], but also in experiments of colloidal systems [13]. More recently, they are considered in theoretical studies of glassy dynamics [14, 15].

Using MD simulations of glassy polymers, we study the detailed dynamics of string-like motions close to the glass transition temperature T_g . Based on the result, we propose how string-like motions contribute to α and β relaxations, and hence derive a theory for glassy dynamics.

II. STRING REPETITION AND INTERACTION

We adopt a bead-spring model of polymer melts in which Lennard-Jones (LJ) particles are connected by nonlinear springs to form chains [16–18] (see Appendix

A for details). Dimensionless LJ units are used in the followings. The T_g of our model is about 0.37 from thermal expansion measurements. At temperature T close to T_g , particle motions consist of thermal vibrations and occasional hops over distances $a \simeq 0.9$, as indicated by peaks in displacement histograms. Figure 1(a)-(b) shows a typical hop of a particle as represented by the true trajectory $\vec{r}(t)$ [19] and the coarse-grained trajectory $\vec{r}^c(t)$ which equals $\vec{r}(t)$ averaged over periods of 5. As seen, the coarse-grained trajectory clearly illustrates the hopping motion and will be adopted throughout this work. In our main simulations, a snapshot of $\vec{r}_i^c(t)$ for every particle i is recorded after every period of 50,000. We consider particle i hopped during a given period if the recorded $\vec{r}_i^c(t)$ shows any displacement beyond a distance of 0.6 within *any* sub-interval of the period.

It is instructive to visually identify string-like motions directly by plotting the trajectories of all the hopped particles in a neighborhood. Figure 1(c) shows a typical string-like motion. A key observation that has motivated this work is that string-like motions become highly repetitive exhibiting strong memory effects at low T . For the string-like motion shown in Fig. 1(c), it is subsequently reversed, repeated and reversed again as shown in Fig. 1(d). There are thus 4 successive string-like motions. String-like motions at higher T are conventionally considered structural, or α relaxations [10]. Observing here that they do not contribute to net displacements, we propose that these oscillatory parts of the motions should be classified only as caging dynamics or β relaxations alongside with thermal vibrations. After repetitions finally stop and assuming an odd number of hops, only the last outstanding hop of each particle contributes to α relaxations. Our simulations show that repetitions become increasingly common and persistent as T decreases. For example, repetitions of more than 15 cycles as well as ones lasting for 5×10^6 are commonly observed at $T = 0.30$. String-like motions which repeat over long durations are usually those isolated from other string-like motions.

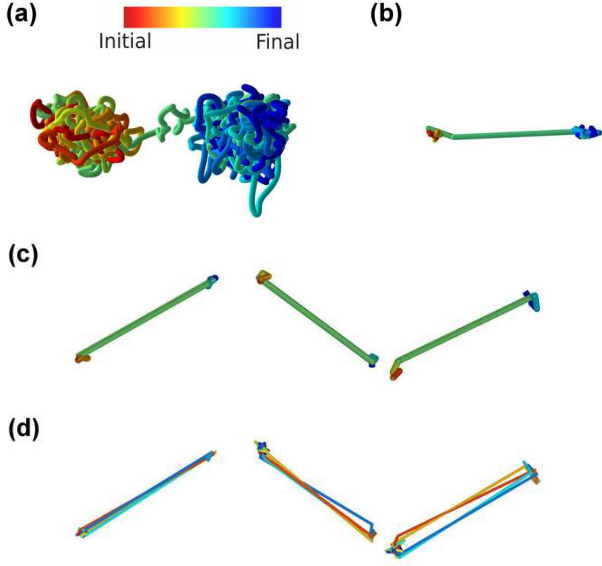


FIG. 1. Particle trajectories colored according to time t with red (blue) being the earliest (latest). (a) Hopping of a single particle as shown by its true trajectory $\vec{r}(t)$. The particle first vibrates about one site and then hops to another site to the right. (b) The same motion as in (a) represented by the coarse-grained trajectory consisting of line segments joining time-averaged positions $\vec{r}^c(t)$. Vibrations before (red) and after (blue) the hop are deemphasized by the averaging. The long line segment (green) implies a sudden large displacement, i.e. a hop. (c) Coarse-grained trajectories of 3 particles arranged in a line performing a string-like motion. Particles hop to the right and nearly fill the positions of the preceding particles. The uniform color (green) of the 3 long segments implies simultaneous hoppings. (d) Coarse-grained trajectories of the same particles in (c) shown for a longer duration. Four string-like motions (red, orange, light blue, blue) occur, the first of which is the one in (c). The durations shown are 200 in (a)-(b), 5,000 in (c) and 20,000 in (d), while $T = 0.32$ in LJ units.

We define a *string* of length n as the set of n particles and a single void which can generate one or more identical or reversed string-like motions. Here, we refer to quasi-equilibrium particle positions like the two shown in Fig. 1(a) as sites and a site is said to be occupied by a void if it is empty. We hence describe Fig. 1(d) as showing 4 related string-like motions generated by a single string. While string-like motions describe the actual dynamics, a string is a configurational feature dictated by the particle energy landscape. As string-like motions are random processes, some strings may never generate any string-like motion during their lifetime. Each string-like motion can involve the single hops of the n particles, or equivalently n hops of the void in the opposite direction. We have analyzed the observed string-like motions following standard algorithms [12] acclimated for shorter observation periods to allow applications to low T . String lengths are found to nicely follow an exponential distribution down to the string length of 1 with an average

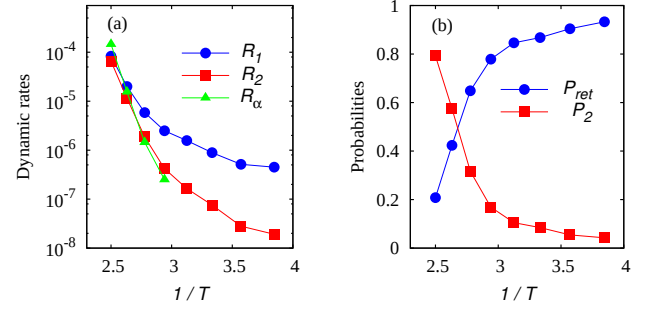


FIG. 2. (a) Plot of particle hopping rate R_1 , rate of non-returning second hop R_2 , and MSD-based α relaxation rate R_α vs $1/T$ for $T = 0.40, 0.38, \dots, 0.26$. (b) Plot of probabilities of returning hop P_{ret} and non-returning second hop P_2 .

string length $l \simeq 2$ varying only mildly with the observation period and T .

To understand string-like oscillations intuitively, we suggest that a void is typically trapped at low T by the rugged energy landscape to within $n + 1$ adjacent sites with similar energies that are lower than those of the surrounding sites. The thermally activated discrete random walks of the void among these lower energy sites are equivalently oscillatory string-like motions of length n . Note that many string-like motions especially the long ones are not fully coherent with particles hopping cooperatively only in micro-strings of length 2 to 3 [11] with occasional partial reversals [11, 20]. The random walks of a void hence comprise short-range correlated steps with a wide distribution of hopping barriers along the path.

Our main analyses focus on particle hopping statistics which are independent of the precise definition of strings. We calculate the particle hopping rate $R_1 = P_1/\delta t$ using the probability P_1 that particle i hops during $[t_0, t_0 + \delta t]$ where $\delta t = 500$. Figure 2(a) plots R_1 against $1/T$. The rate of string like-motions per site is simply R_1/l . We also study the subsequent motion of a hopped particle i by evaluating the probabilities P_{ret} and P_2 that it will first perform, respectively, a returning or non-returning second hop within the 10^6 duration after the first hop. A long duration of 10^6 is used so that undetermined cases occur at a small probability of $1 - P_{ret} - P_2 \lesssim 0.05$. Here, a hop is defined as returning if the particle returns to within a distance of 0.3 from the original position $\vec{r}_i^c(t_0)$. A non-returning second hop means that it has displaced again beyond a distance of 0.6 from the hopped position $\vec{r}_i^c(t_0 + \delta t)$. Figure 2(b) plots P_{ret} and P_2 against $1/T$. As T decreases, P_{ret} rises while P_2 drops implying increasing memory effects. Figure 2(a) also plots the rate $R_2 = R_1 P_2$ of non-returning second hops and $R_\alpha = 1/\tau_\alpha$, a measure we use to approximate the α relaxation rate based on the particle mean square displacement (MSD) [21]. Here, τ_α is the time it takes for the MSD to reach $(0.6)^2$, a value compatible with our definition of hops for easy comparison.

For $T \geq 0.38 > T_g$, Fig. 2(a) reveals $R_1 \simeq R_2 \simeq R_\alpha$

implying that most hops are α relaxations. However, for $T_g > 0.36 \geq T \geq 0.34$, $R_1 \gg R_\alpha$ showing that particle hops, which have become increasingly repetitive (as indicated by P_{ret} in Fig. 2(b)), are more frequent than the rate of α relaxations. This supports the assertion that oscillatory hops are β relaxations. On the other hand, $R_2 \simeq R_\alpha$ still holds. This motivates us to propose that second non-returning hops now dictate the α relaxation rate. Note that our samples are not fully thermalized at $T \lesssim 0.36$. Nevertheless, the observation $R_1 \gg R_2 \simeq R_\alpha$, which is based on comparisons between values that are affected similarly by the same non-equilibrium effects, should be robust. We have also checked that values admit little change if data from only the second half of the simulations are used.

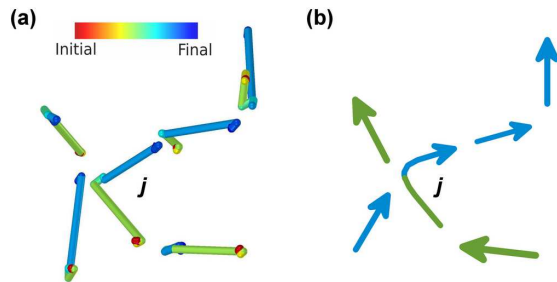


FIG. 3. Displacement pathways of 6 particles during a pair-interaction between two strings illustrated using (a) coarse-grained trajectories and (b) the corresponding schematic diagram. Trajectories in (a) are colored according to time and the two distant colors (green and blue) of the 7 long segments imply two set of hoppings performed at different times. The first string (green) involves the hopping of 3 particles. It pushes particle j to a new position along the second string (blue) whereby the 4 particles it contain can hop subsequently. Particle j is shared by both strings and has hopped twice while each of the other 5 particles have hopped only once during the observation period of 4500. The temperature is 0.32.

We have visually examined the detailed particle trajectories of dozens of non-returning second hops. Interpretation for their origin get significantly easier at low T . For $T \lesssim 0.32$, all of them can be identified as being caused by pair-interactions of strings similar to the example of string crossing shown in Fig. 3(a)-(b). Here, a particle labeled as j is shared by two strings. The first string-like motion induces the first hop of particle j . The new position of j enables a new string. The second string-like motion along the new string then induces the second hop of particle j . This however disable the first string so that it cannot be reversed unless the second string is first reversed. If not reversed, all 7 hops contribute to α relaxations.

Based on these observations, we propose that without pair interactions, string-like oscillations occur indefinitely exhibiting memory effects and contribute only to β relaxations. Pair interactions can break the repetition and generates α relaxations. Since each string much have a void, pair interactions of strings are equivalently pair

interactions of voids. Furthermore, pair interactions can also be reversed. We have observed reversal and repetition of ordered sequences of string-like motions. They generalize simple string-like oscillations and exemplify memory effects at longer time scales. Thus, pair interactions only generate α relaxations if they are not eventually reversed by memory effects.

III. THEORY OF INTERACTING STRINGS

We now propose a minimal theory of glassy dynamics based on the evolution of strings under pair-interactions in the form of string crossings. Assume for simplicity that the site which is energetically most stable for the void is at one end of the string. We refer to this site as the head of the string. Let γ_e be the rate at which the void is excited from the head initiating a forward string-like motion. Suppose the void reaches the other end and remains there for an average period of τ_l , which we refer to be the lifetime of an excited string. Afterward, the void returns to the head producing a backward string-like motion unless string interactions take place. To simplify the algebra, we assume that an individual string-like motion is fast compared with τ_l and always transverses the whole string. The probability that a string is excited is hence $\tau_l \gamma_e$.

A pair interaction requires that two strings cross at a common site. We classify each newly created string as being *active* if it coincides spatially with another existing string, or as *passive* if it is isolated. Let $S = S_A + S_P$ be the total density of strings present in the system and S_A (S_P) be the density of active (passive) strings. Our theory is of the mean field type based on S and S_A , which provide the simplest measure of the spatial correlation of the strings. Let κ be the probability that a new string is classified as active. To estimate κ , consider partitioning the space into independent boxes each of volume $V_s = l^3 \Omega$, where Ω is the volume of a particle. Assume that each box can hold an isolated string or multiple overlapping strings and new strings are randomly assigned to boxes, we get $\kappa = 1 - \exp(-SV_s)$ using the Poisson distribution.

Since active strings interact with both active and passive strings while passive strings do not interact among themselves, we adopt a pair interaction rate density (see Appendix B for details)

$$\chi = \frac{1}{2} \alpha S_A^2 + \alpha S_P S_A \quad (1)$$

where $\alpha = \tau_l \gamma_e^2 V_s$ based on simple chemical kinetics for the allowed interactions. After an interaction, both voids of the strings are relocated as can be inferred from Fig. 3. For simplicity, we truncate all memories after a pair-interaction which amounts to assuming that the new void positions become the heads of two completely independent new strings. All particles in both strings then contribute to uncaging hops and this gives an α relaxation

rate

$$R_\alpha = 2l\chi\Omega \quad (2)$$

Moreover, new strings are enabled at a rate density 2χ , each of which has a probability κ to be active. Simultaneously, active strings are disabled at a rate density αSS_A according to Eq. (1). Thus, $\dot{S}_A = 2\kappa\chi - \alpha SS_A$, which gives

$$\dot{S}_A = -\alpha\kappa S_A^2 + \alpha(2\kappa - 1)SS_A \quad (3)$$

Ideal Transition: We first derive an ideal theory describing an ideal (i.e. true) phase transition between two distinct phases. Equation (3) is formally the logistic equation for population dynamics [22]. It exhibits a continuous bifurcation transition at $\kappa = 1/2$ between a die-out and a stable phase which correspond here to a mobile and a frozen phase. Assuming a string density $S = \tilde{S} \exp(-E_S/kT)$ where E_S is the free energy cost of a void and $\tilde{S} = \Omega^{-1}$, the transition then occurs at an ideal critical temperature given by

$$T_c = \frac{E_S}{k \ln(\tilde{S}V_s / \ln 2)} \quad (4)$$

This is because at equilibrium, $\dot{S}_A = 0$ and Eq. (3) reduces to $-\alpha\kappa S_A^2 + \alpha(2\kappa - 1)SS_A = 0$. Below T_c (corresponding to $\kappa < 1/2$), Eq. (3) has a single trivial equilibrium solution $S_A = 0$. It represents the frozen phase as Eqs. (1)-(2) implies $R_\alpha = \chi = 0$. In contrast, above T_c corresponding to $\kappa > 1/2$, $S_A = 0$ becomes an unstable solution and is irrelevant. A new stable solution $S_A = (2 - 1/\kappa)S > 0$ emerges and represents a mobile phase with $R_\alpha \propto \chi > 0$. Combining the two cases,

$$S_A = \max\{0, (2 - 1/\kappa)S\} \quad (5)$$

Intuitively, the criterion $\kappa > 1/2$ for the mobile phase means that at least half of the new strings must be active to give sustainable pair-interactions as each interaction needs at least one active string.

Assuming Arrhenius forms for the string density, excitation rate and lifetime, i.e. $S = \tilde{S} \exp(-E_S/kT)$, $\gamma_e = \tilde{\gamma}_e \exp(-E_e/kT)$ and $\tau_l = \tilde{\tau}_l \exp(E_l/kT)$ with $\tilde{\tau}_l = 0.5/\tilde{\gamma}_e$, R_α can be calculated using Eqs. (1), (2) and (5). An application to an viscosity calculation will be presented below. At this point, we note that similar to other theories for ideal transitions [4], R_α drops too abruptly to 0 as T is decreased to T_c and this disagrees with the fact that glass transitions occur over extended temperature ranges.

Shifted Transition: The above assumption of a uniform string excitation rate γ_e is unrealistic and a broad distribution is expected. In principle, each interstitial site can be squeezed in with a particle and is potentially the head of a string with albeit a small γ_e in most cases. When T is decreased towards T_c calculated at a given $\tilde{S}$, the predicted dynamics slow down due to proximity to the underlining ideal transition to the frozen phase. However, additional strings with smaller values of

γ_e comparable to the rate of the slowed-down dynamics should then be considered. This decreases T_c according to Eq. (4) and the system remains slightly above the ideal transition for an extended temperature range. The ideal transition can in principle be shifted to $T_c \rightarrow 0$ as $T \rightarrow 0$. In fact, there exists a well-known analog in fluid flow in porous media with broad channel width distributions. For similar reasons, fluids typically flow slightly above the percolation threshold with the flow limited by the narrowest considered channels [23].

Quantitatively, we generalize the simple Arrhenius forms to

$$S = (1 + \epsilon)\tilde{S} \exp(-E_S/kT) \quad (6)$$

$$\gamma_e = \tilde{\gamma}_e \exp[-(E_e + \epsilon kT_e)/kT] \quad (7)$$

where ϵ is a parameter for including additional slower strings and T_e is a constant. Also, γ_e now represents the excitation rate of the slowest considered strings that limit the dynamics. The linear dependence of the energy barrier on ϵ in Eq. (7) corresponds to a constant density of state with respect to the string excitation energy. We determine ϵ variationally by numerically maximizing χ . This is because a small ϵ will bring the system too close to the frozen phase predicting too little dynamics, while a large ϵ also decreases χ due to an underestimated rate-limiting γ_e . In addition, we take $T_e \gg T_c$ so that at $T = T_e$, $S \simeq S_A$ and χ is maximized conveniently at $\epsilon \simeq 0$.

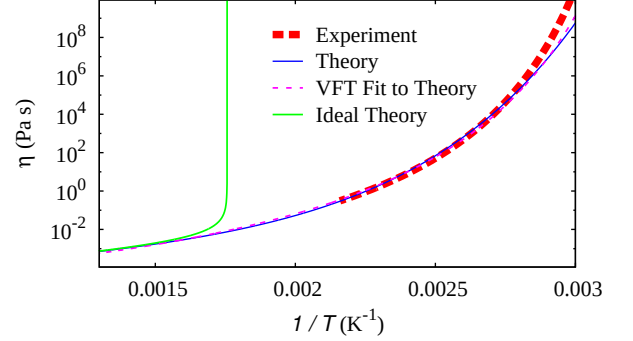


FIG. 4. Plot of viscosity η of polystyrene from experiment [24] (red dashed line), theory (blue solid line), a VFT fit to the theory (purple dashed line), and the ideal theory (green solid line).

Viscosity Calculation: We apply our theory to account for the viscosity of a common glassy polymer, polystyrene (PS). We choose a low molecular weight $M = 2.4$ kg/mol for which Rouse dynamics is fully applicable [25], but generalization to reptation dynamics should be straightforward. Each chain has 23 monomers, each of which is identified as a particle. The Kuhn length of PS is 15 Å corresponding to about 10 monomers [26] so that the number of Kuhn segments in a chain is $N_K = 2.3$. Assuming that a Kuhn segment admits an

α relaxation when any one of its 10 monomers performs an uncaging hop, the relaxation rate of a Kuhn segment is $10R_\alpha$. The polymer relaxation time τ_R according to the Rouse model is N_k^2 times that of a Kuhn segment [25] and thus $\tau_R = 0.1N_k^2/R_\alpha$. The polymer viscosity is given by [25]

$$\eta = \frac{\pi^2}{12} \left(\frac{\rho RT}{M} \right) \tau_R \quad (8)$$

where $\rho = 1.05 \text{ g/cm}^3$ is the density of PS and $R = 8.31 \text{ J/mol-K}$.

To apply to our model, we calculate χ for each value of T considered using Eqs. (1),(5)-(7) which involves maximizing χ with respect to ϵ . We then obtain R_α from Eq. (2) and hence η from Eq. (8). As shown in Fig. 4, we reproduce the experimental values of η from Ref. [24] reasonably well by taking $E_s = 0.12 \text{ eV}$, $E_e = 0.11 \text{ eV}$, $E_l = 0.105 \text{ eV}$, $\tilde{\gamma}_e = 10^{12} \text{ s}^{-1}$, and $T_e = 500^\circ \text{ C}$. As T decreases, ϵ varies from 0.16 to 4.89. Figure 4 also shows extrapolated values to $T > 463 \text{ K}$. We observe that η from our theory can be fitted very well by the Vogel-Fulcher-Tammann (VFT) empirical form with a Kauzmann temperature $T_0 = -1^\circ \text{ C}$. We have also plotted results from the ideal theory which shows a true transition at $T_c = 296^\circ \text{ C}$. Finally, we note that our present equations cannot provide a large fragility and underestimates η at low T as observable in Fig. 4. By generalizing ϵkT_e in Eq. (7) to $\epsilon^{1+\zeta} kT_e$ with $\zeta \geq 0$ corresponding to a non-linear density of states of the strings, a better fit with a higher fragility can be obtained.

IV. CONCLUSIONS

In summary, string-like motions in bead-spring polymer melts are found to be highly repetitive at low temperatures. The repetitive parts of the motions are suggested to be β relaxations. We propose that α relaxations are dominated by the disruptions to the repetitive string-like motions by pair-interactions among strings. We hence derive a theory of glassy dynamics which can reproduce the viscosity of a common polymer glass and the VFT empirical form.

In view of the rich ramifications of glassy dynamics, there are numerous possible additional studies that can be carried out to further scrutinize the theory. We have extended the theory to a spatial dependent form and applied it to thin films with a free surface. This allows the explanations of a surface mobile layer previously observed in experiments [27–29] and an exponential depth-dependent flow profile found in simulations [18]. Results will be reported elsewhere. Possible future works may include the extension of the MD study to other binary or monoatomic glassy models, although we expect similar results as string-like motions are remarkably similar in various systems [9, 11, 12]. It will also be interesting to consider memory effects beyond the first pair-interaction and relate them to the memory kernel of the

mode-coupling theory [4]. We have analyzed the dynamics using a mean field theory. We expect that a full description should be of the reaction-diffusion type with a limited diffusion range, which are known to generate non-trivial spatio-temporal patterns in chemical, biological and other complex systems [30]. The spatio-temporal correlations may explain the dynamic heterogeneity observed in glasses.

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Appendix A: Model and simulation method

Our simulations based on a bead-spring model of polymer follow Refs. [16–18] generalized to heavier chain-tails for producing a uniform monomer mobility. Specifically, pairs of particles interact via the Lennard Jones (LJ) potential $4\epsilon \left[(\sigma/r)^{12} - (\sigma/r)^6 \right]$ with a cutoff $2 \cdot 2^{1/6}$ beyond which it becomes a constant. The potential hence has both attractive and repulsive parts. Bonded particles are further bounded by a finitely extensible nonlinear elastic (FENE) potential $-\frac{k}{2} R_0^2 \ln \left[1 - (r/R_0)^2 \right]$ where $k = 30\epsilon/\sigma^2$ and $R_0 = 1.5\sigma$. We adopt LJ units which amounts to taking $\sigma = \epsilon = 1$. Internal particles in each chain have a mass $M_I = 1$. However, in contrast to common practice [16–18], one particle at each end of the chain has a larger mass $M_T = 4$. This leads to a uniform mobility for all particles in the chain as is verified from measured particle mean square displacements.

All simulations are executed using HOOMD [31] on Nvidia GTX580 GPU's. Our main simulations involve two samples of polymer melts each having 1500 chains of length 10 inside a cubic box with periodic boundary conditions simulated using a time step of 0.005. The melts are first thermalized at $T = 0.6$ and then 0.5 following standard techniques [16]. They are then repeatedly quench and anneal under NPT conditions at zero pressure by steps of $\Delta T = 0.02$. Each of these quenching or annealing processes involves 10^7 time steps. At each T studied, the samples are further annealed for 10^8 time steps before data taking during 10^9 additional time steps under NVT conditions. The data includes 10^4 snapshots of both instantaneous particle positions $\vec{r}_i(t)$ and coarse-grained positions $\vec{r}_i^c(t)$ taken after every 10^5 time steps. Each recorded value of $\vec{r}_i^c(t)$ is itself an in-situ average over 20 instantaneous positions of particle i taken after every 50 time steps.

Appendix B: Pair interaction rate

In our model, strings are basically immobile except for a single hop by each particle. A string migrates over a limited range only after an interaction. The evolution of

strings under pair interactions is analogous to the chemical kinetics of immobile reactants such as gas molecules adsorbed on solid surfaces [30]. It is tempting to assume a pair interaction rate density $\chi = \frac{1}{2}\alpha S^2$ based on simple chemical kinetics for well-mixed reactants. However, lacking any information about spatial correlations in the quenched string positions, it cannot account for the important situation that at a small but finite $S > 0$, the strings can evolve into a mutually isolated frozen state so that $\chi = 0$ exactly. We hence define in the main text S_A and S_P which minimally keep track of the quenched disorders. The ratio S_A/S_P is a measure of the state of spatial overlapping of the strings. Noting that active strings react with both active and passive strings, we adopt

$$\chi = \frac{1}{2}\alpha S_A^2 + \alpha S_P S_A. \quad (\text{B1})$$

Here, a possible $\frac{1}{2}\alpha S_P^2$ term is omitted because passive strings do not react with each other. With the quenched disorder minimally accounted for by the omitted term, the allowed interactions are now described based on simple chemical kinetics of binary reactions for well-mixed reactants.

To calculate α , we first consider the case of a large S corresponding to a high T . Each independent box of volume V_s (see main text for definition) is occupied on average by $N_s = SV_s$ strings with $N_s \gg 1$ so that all strings are active (i.e. $S = S_A$). Inside the box, all strings are assumed to overlap spatially and ready to interact.

A pair interaction also requires the simultaneous excitation of both strings. During a period $\tau_e = 1/\gamma_e$, each string on average is excited once. The period consists of $m = \tau_e/\tau_l$ subintervals each equals the lifetime τ_l of the excited string. The chance that any two of the N_s strings are excited at the same subinterval is $1/m$. The total interaction rate in a box is hence $R_I = (1/2)N_s^2/m\tau_e$. This gives an interaction rate density $\chi = R_I/V_s$ which can be simplified to

$$\chi = \frac{1}{2}\tau_l\gamma_e^2 V_s S_A^2 \quad (\text{B2})$$

A comparison with Eq. (B1) leads to

$$\alpha = \tau_l\gamma_e^2 V_s \quad (\text{B3})$$

after noting $S_P = 0$.

As a consistency check, we consider the fluid phase close to criticality with $S_A \ll S_P$, which is most relevant to our work. Then, active strings are few in number and most of them must interact with passive strings. An independent box contains an active string with a probability $S_A V_s$. Upon classified as an active string, the box must already hold a passive string. Similar to the calculation above, considering also the simultaneous excitation of both strings gives an interaction rate $R_I = \tau_l\gamma_e^2 S_A V_s$ and hence a rate density $\chi = \tau_l\gamma_e^2 S_A$. A comparison with Eq. (B1) gives $\alpha = \tau_l\gamma_e^2/S_P$, which can be rewritten as $\alpha = \tau_l\gamma_e^2 V_s/\kappa$ using $\kappa \simeq SV_s \simeq S_P V_s$. It differs to Eq. (B3) only by a factor $1/\kappa$ which equals 2 at criticality. For simplicity, we adopt Eq. (B3) by neglecting this factor which gives no qualitative change to our results.

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